

Owned Requirements: FR-009 to FR-011, FR-015 to FR-018

Owned Components: Category/Assignment Module, Search/Tracking Module

Specialist Section: None beyond test cases

## 13. Test Case Management & Traceability

### 13.1 Test Cases: Category/Assignment Module

| Test Case ID | Requirement ID | Description                                   | Preconditions                           | Steps                                                                | Expected Result                                   | Type     |
|--------------|----------------|-----------------------------------------------|-----------------------------------------|----------------------------------------------------------------------|---------------------------------------------------|----------|
| TC-ASG-01    | FR-009         | List predefined categories                    | Categories seeded; user logged in       | GET /api/categories                                                  | 200 with the seeded category list                 | Positive |
| TC-ASG-02    | FR-009         | Submit complaint with a non-existent category | Resident logged in                      | Submit a complaint with an unknown category_id                       | 400/404; complaint is not created                 | Negative |
| TC-ASG-03    | FR-010         | Admin creates a category                      | Admin logged in                         | POST /api/categories with name "Plumbing"                            | 201; category appears in the list                 | Positive |
| TC-ASG-04    | FR-010         | Resident tries to create a category           | Resident logged in                      | POST /api/categories                                                 | 403; no category created                          | Negative |
| TC-ASG-05    | FR-010         | Create category with duplicate name           | "Plumbing" exists; admin logged in      | POST /api/categories with the same name                              | 409; no duplicate row created                     | Negative |
| TC-ASG-06    | FR-011         | Admin assigns complaint to staff              | Complaint is Submitted; admin logged in | POST /api/complaints/{id}/assignment with valid department and staff | 200; status becomes Assigned; history row written | Positive |
| TC-ASG-07    | FR-011         | Non-admin attempts assignment                 | Staff or resident logged in             | POST /api/complaints/{id}/assignment                                 | 403; status unchanged                             | Negative |
| TC-ASG-08    | FR-011         | Assign a Closed complaint                     | Complaint is Closed; admin logged in    | POST /api/complaints/{id}/assignment                                 | 409; no new assignment created                    | Negative |

## 13.2 Test Cases: Search/Tracking Module

| Test Case ID | Requirement ID | Description                                 | Preconditions                                         | Steps                                             | Expected Result                                  | Type     |
|--------------|----------------|---------------------------------------------|-------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|----------|
| TC-TRK-01    | FR-015         | Resident views own complaint status         | Resident has a complaint                              | GET /api/complaints/{id}/status                   | 200 with the current status                      | Positive |
| TC-TRK-02    | FR-015         | Resident views another resident's complaint | Complaint belongs to resident B; resident A logged in | GET /api/complaints/{id}/status                   | 403/404; no data exposed                         | Negative |
| TC-TRK-03    | FR-016         | History records status changes in order     | Complaint moved Submitted, Assigned, In Progress      | GET /api/complaints/{id}/history                  | 3 entries in time order with actor and timestamp | Positive |
| TC-TRK-04    | FR-016         | Attempt to modify or delete history         | Any user logged in                                    | PUT/DELETE /api/complaints/{id}/history           | 405/403; history unchanged                       | Negative |
| TC-TRK-05    | FR-017         | Search by Complaint ID                      | Complaint exists; user logged in                      | GET /api/complaints?q=<complaint_id>              | 200 with that complaint                          | Positive |
| TC-TRK-06    | FR-017         | Search with SQL injection string            | User logged in                                        | GET /api/complaints?q=' OR 1=1 --                 | 200 with empty list; no error; no extra data     | Negative |
| TC-TRK-07    | FR-018         | Filter by status and category               | Mixed complaints exist                                | GET /api/complaints?status=Assigned&category_id=X | Only matching complaints returned                | Positive |
| TC-TRK-08    | FR-018         | Filter with invalid status value            | User logged in                                        | GET /api/complaints?status=Foo                    | 400 invalid filter                               | Negative |
| TC-TRK-09    | FR-017         | Resident search is scoped to own complaints | Complaints from two residents match the term          | Resident A searches a term matching both          | Only A's complaints returned                     | Negative |

## 13.3 Requirement Traceability

| Requirement ID | Component                  | Test Case IDs                   |
|----------------|----------------------------|---------------------------------|
| FR-009         | Category/Assignment Module | TC-ASG-01, TC-ASG-02            |
| FR-010         | Category/Assignment Module | TC-ASG-03, TC-ASG-04, TC-ASG-05 |
| FR-011         | Category/Assignment Module | TC-ASG-06, TC-ASG-07, TC-ASG-08 |

| Requirement ID | Component              | Test Case IDs                   |
|----------------|------------------------|---------------------------------|
| FR-015         | Search/Tracking Module | TC-TRK-01, TC-TRK-02            |
| FR-016         | Search/Tracking Module | TC-TRK-03, TC-TRK-04            |
| FR-017         | Search/Tracking Module | TC-TRK-05, TC-TRK-06, TC-TRK-09 |
| FR-018         | Search/Tracking Module | TC-TRK-07, TC-TRK-08            |

### 13.4 Test Data / Evidence to Collect

- Seeded category list and API response for category listing.
- Evidence of admin-only category creation and rejection for residents.
- Evidence of successful assignment, the resulting status change and the history entry.
- Evidence that assignment of Closed complaints is rejected.
- Evidence that residents cannot view other residents' complaints.
- Status history output showing ordered entries with actor and timestamp.
- Search and filter results for valid, invalid and injection-style inputs.